DirectGCode
AI CAM Technical Thesis

Executive Edition

Independent technical research on controlled AI-assisted CAM workflows for CNC programming.

Author: Lorenzo Marsico

Publication package: public, professional, non-commercial, non-proprietary.

Publication Scope

- Independent technical research and public thesis edition.
- No proprietary source code, internal algorithms, full datasets, complete knowledge packs, customer data, or confidential project information are disclosed.
- AI recommendations discussed here are framed as decision support, not autonomous release of CNC programs.

Executive Summary

DirectGCode explores an AI CAM architecture for CNC programming in which artificial intelligence supports technical decisions without replacing the CAM programmer or bypassing industrial validation. The thesis focuses on a controlled workflow: technical input, case retrieval, AI-assisted proposal, operator review, deterministic CAM generation, validation, and shop-floor output.

The central principle is not blind automation. The useful industrial pattern is an assisted system that can retrieve relevant cases, suggest tools and parameters, expose warnings, retain validated know-how, and keep the human operator in control of critical decisions.

Industrial Context

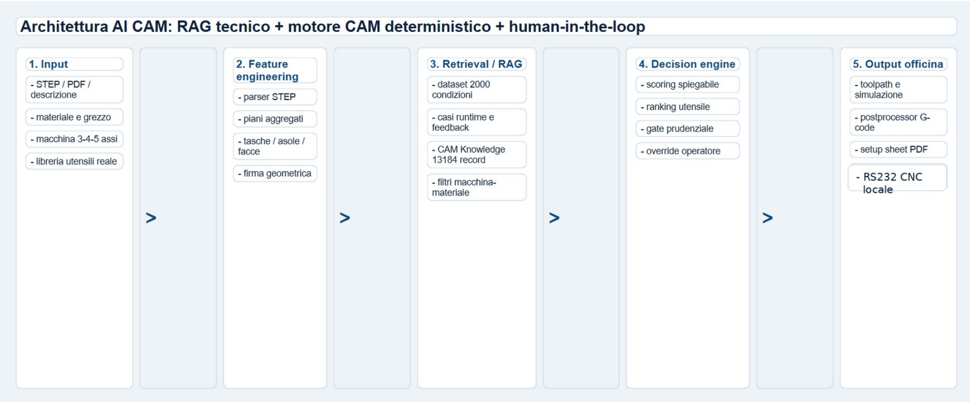
CNC programming is a chain of practical decisions. Material, geometry, setup rigidity, tool availability, machine constraints, quality requirements, cycle time, and operator experience all shape the final program. A professional AI CAM system must therefore be explainable, local-first where confidentiality matters, and designed around traceability.

Objectives

- Reduce repetitive CAM preparation work while preserving expert review.
- Make shop-floor know-how more structured, reusable, and auditable.
- Separate AI suggestions from deterministic generation and validation.
- Support local operation for industrial continuity and confidentiality.
- Use controlled learning rather than uncontrolled self-modification.

General Architecture

The architecture separates user interface, AI/CAM services, deterministic CAM core, local knowledge storage, validation, postprocessing, and optional headless/API components. This separation allows the project to evolve without exposing source code or internal implementation details.



High-level DirectGCode AI CAM architecture extracted from the public thesis.

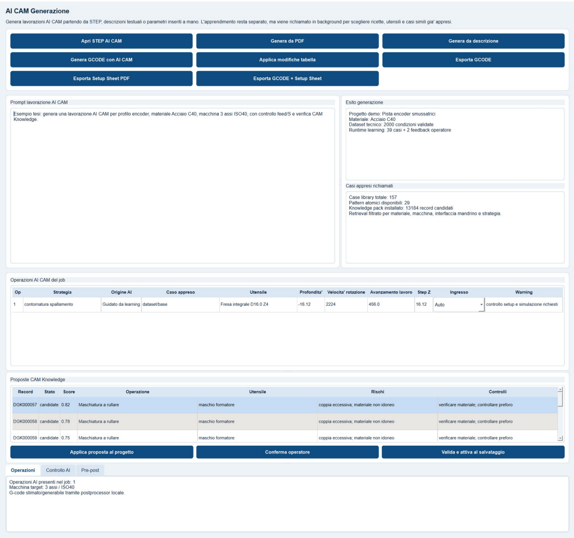
Human-In-The-Loop Model

The Human-In-The-Loop model is treated as a production requirement. AI output remains a proposal until the operator reviews it. This matters because CNC output has physical consequences: a weak recommendation can become tool damage, part scrap, or machine downtime if it is released without control.

- The operator can accept, edit, reject, or override recommendations.
- Warning signals are part of the value delivered by the system.
- Critical outputs require validation gates before export.

AI CAM Workflow

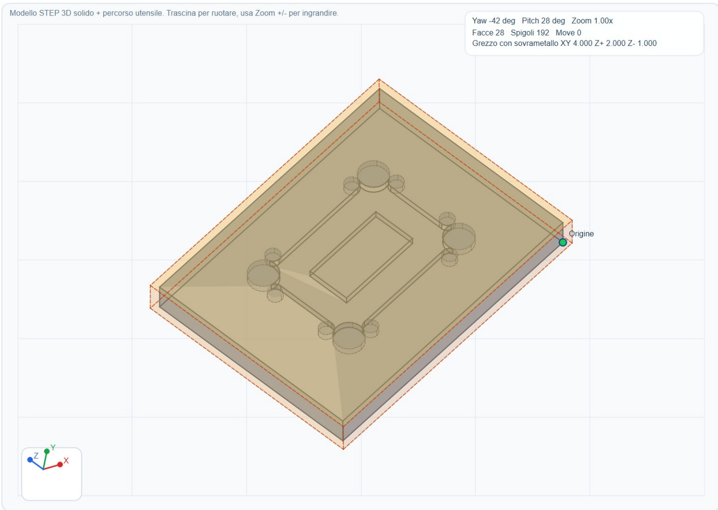
The AI CAM layer can combine feature engineering, case retrieval, scoring, ranking, parameter suggestions, warnings, and fallback heuristics. It is better understood as a composed decision-support system than as a single black-box model.



AI CAM generation panel and review workflow from the thesis.

STEP And Toolpath Visualization

Geometry and toolpath visualization help the operator inspect machining intent before output generation. This visual layer supports trust because it turns an abstract recommendation into something the programmer can reason about.



STEP viewer example used in the public thesis.

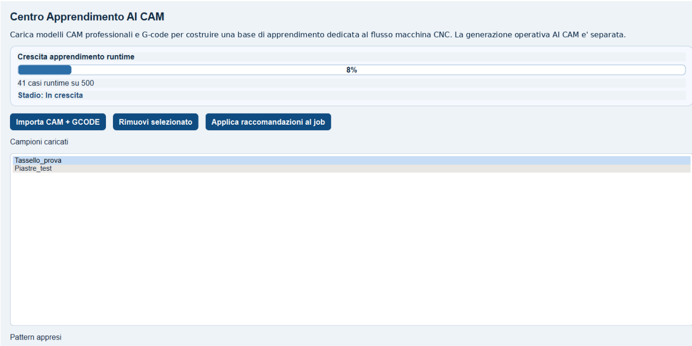
Industrial RAG

DirectGCode adapts retrieval-augmented generation to the CAM domain. Instead of retrieving only text, the system can retrieve comparable technical cases, machining records, validated patterns, warnings, and conditions that match the current job context.

For industrial use, retrieval is valuable only if it is explainable: why a case is relevant, which constraints match, which risks apply, and whether the record has been validated by an operator or shop-floor result.

Controlled Learning

Learning is framed as knowledge retention. The thesis separates base datasets, runtime cases, operator feedback, imported professional samples, and candidate knowledge records. This avoids treating every new event as immediate production truth.



Controlled learning panel and knowledge retention workflow.

AI Governance

- Local-first workflows protect sensitive production data and continuity.
- Feature flags allow gradual activation of new capabilities.
- Shadow mode permits recommendation testing before operational authority.
- Memory separation reduces contamination between base knowledge, local feedback, and candidate records.
- Versioning and rollback are necessary for code, models, knowledge packs, and configuration.

Roadmap

- Extend validation with more real industrial jobs, materials, and machine profiles.
- Improve simulation, collision checks, and machine-envelope verification.
- Containerize only separable services while preserving the desktop production workflow.
- Strengthen explainability of recommendations, warnings, and case retrieval.
- Build safer update and rollback processes for models and knowledge components.

Publication Scope

This Executive Edition is part of a public GitHub publication package. It intentionally avoids source code, proprietary algorithms, full datasets, knowledge pack contents, customer files, and confidential operational material. The goal is to communicate the technical architecture, industrial reasoning, and governance model in a professional and shareable form.